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RESEARCH ARTICLE

Assessment of Greenhouse Gas Emissions and Mitigation Strategies for a University in Bangladesh

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ABSTRACT Climate change is an urgent global challenge. Higher education institutions (HEIs) have a crucial role in addressing this issue by reducing their environmental impact. This study presents the first comprehensive assessment of greenhouse gas (GHG) emissions at a university in Bangladesh using the GHG Protocol Corporate Standard to categorize emissions into scopes 1, 2, and 3. The total GHG emissions amount to 2498.80 tons of carbon dioxide equivalent (tCO_2e) annually, with a per capita carbon footprint of 0.4671 tCO_2e. Scope 1 emissions, primarily from university vehicles, generators, cafeterias, and air conditioners, account for 29.73% (742.55 tCO_2e). Scope 2 emissions, related to purchased electricity from power grid, are at 18.18% (454.14 tCO_2e). The largest contributor is the indirect emissions from commuting in scope 3, accounting for 52.10% (1302.14 tCO_2e) of the total. Major emission sources include air conditioners, purchased electricity and commuting of university personnel by cars. To reduce these emissions, the study presents a scenario based mitigation framework consisting of Conventional, Incremental, and Comprehensive Scenarios with targeted interventions for each scope. These include addressing high fugitive emissions from airconditioner by replacing current coolant gases with more efficient alternatives such as R32, transitioning to hybrid or solar-powered electric vehicles, expanding shared transport options, and integrating rooftop and floating solar systems. Although the study is based on one university, the proposed strategies are designed to be broadly applicable to universities across Bangladesh. The study also emphasizes energy efficiency, eco-friendly mobility, and integrating sustainability into the curriculum. This study underscores the need for localized assessments and tailored strategies to effectively address the unique challenges of reducing GHG emissions in HEIs, providing a robust framework for creating a more sustainable university community.

INDEX TERMS Carbon footprint, greenhouse gas emissions, GHG protocol, mitigation strategies, university campus.

I. INTRODUCTION

Climate change, driven by rising greenhouse gas (GHG) emissions, is one of the most urgent crises of our era, affecting ecosystems, economies, and communities worldwide. Higher education institutions (HEIs)—as hubs of innovation and knowledge—are uniquely positioned to contribute to climate action by reducing their own carbon footprints and promoting sustainability-focused education and research. The United Nations' Sustainable Development Goals (SDGs) [1] provide a holistic global framework for addressing interconnected societal and environmental challenges. Among

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these objectives, SDG 13 -"*Take urgent action to combat climate change and its impacts*" - emphasizes the severity of the climate crisis and the importance for immediate actions to reduce its detrimental effects. While governments and industries play significant roles in achieving SDG targets, HEIs are also key stakeholders in climate action. Universities not only shape future leaders and policymakers but also operate large facilities that consume considerable energy, generate commuting demands, and utilize diverse resources. Addressing these emissions is vital for universities to align with global sustainability goals. While SDG 13 calls for urgent action on climate change, SDG 7 - "*Ensure access to affordable, reliable, sustainable, and modern energy for all*" – is equally important. In the context of climate change, this goal emphasizes the need to transition to cleaner, more sustainable energy sources. Universities can contribute significantly by adopting clean energy usages, reducing carbon footprints through renewables, and promoting energy efficiency. Thus, universities can play a vital role in advancing integrated climate action and public awareness around sustainable energy solutions.

Despite growing awareness of climate change's far-reaching impacts, many universities lack a standardized approach to measuring and reducing their carbon footprint. A robust framework is essential for accurate emissions tracking and for developing targeted mitigation strategies. The GHG Protocol Corporate Standard, established by World Resources Institute (WRI) and World Business Council for Sustainable Development (WBCSD) [2], provides a globally recognized methodology. This framework categorizes emissions into scopes 1, 2, and 3, which helps universities to identify key areas for improvement. The GHG Protocol Corporate Standard by WRI and WBCSD [2], a widely recognized and adopted standard for GHG accounting and management, provides the essential structured methodology and guidelines for institutions and organizations to assess and reduce their carbon footprints comprehensively. The GHG Protocol classifies emissions into three distinct scopes, each representing different sources of emissions. Scope 1 emissions are the direct emissions from sources owned or controlled by the organization, for example, emissions from on-campus fossil fuel consumption and university-owned vehicles. Scope 2 emissions are the indirect emissions from off-campus energy sources used to generate the electricity purchased by the organization. Scope 3 emissions cover other indirect emissions, which are typically more challenging to calculate and involve activities such as commuting, waste management, and supply chain-related emissions. For universities, scope 3 emissions often include a wide range of activities beyond their direct control but still connected to their operations and activities. By adopting this framework, HEIs can measure their GHG emission and take effective actions to reduce their carbon footprint and thus contribute significantly to the global mission of combatting climate change.

While universities in developed countries have made strides in reducing their carbon footprints, HEIs in developing nations continue to face significant challenges. Bangladesh, one of the most climate-vulnerable countries, has an urban landscape with rising emissions from fossil-fuel-dependent electricity and inefficient transportation systems. Situated in the heart of Dhaka, the University of Liberal Arts Bangladesh (ULAB) exemplifies these constraints through its heavy reliance on air conditioning, high commuting-related emissions, and dependence on fossil fuels. These patterns reflect common challenges faced by institutions across South and Southeast Asia, underscoring the global relevance of a localized case study. It faces unique challenges, including limited access to renewable energy, high commuting-related emissions, and reliance on fossil fuels. By focusing on Scopes 1, 2, and 3 emissions, this paper not only assesses ULAB's emissions profile but also identifies practical strategies for GHG reduction which align with Bangladesh's national commitments to the SDGs and contribute to global efforts in combating climate change. Although air quality concerns such as $PM_{2.5}$ and PM_{10}, along with resource efficiency, are relevant to the broader environmental impact of university operations, this study is limited to GHG emissions in accordance with the GHG Protocol Corporate Standard.

To understand how HEIs are tackling GHG emissions globally, this paper reviews prior studies from North and South America, Europe, the Middle East, and Asia. These studies highlight the diverse challenges faced by universities worldwide.

In the United States, HEIs have made significant efforts to assess and manage their GHG emissions. Sinha et al. [3] examined the GHG emissions from U.S. institutions of higher education participating in the American College and University Presidents' Climate Commitment (ACUPCC). Purchased electricity, stationary combustion, and commuting were identified as emission sources of 88% of total emissions. Klein-Banai and Theis [4] analyzed GHG emissions in 135 U.S. colleges and universities, highlighting the complexity of emissions due to factors like institutional size, laboratory space, residential areas, and commuting distances. They found that laboratory space has ten times the emissions impact per square meter compared to other areas, indicating greater challenges for institutions with lab-intensive curricula in reducing their GHG footprint.

GHG assessments in Latin American and North American universities highlight high Scope 3 emissions from commuting and energy use, with limited focus on mitigation strategies. A study conducted at Curico Campus, Universidad de Talca in Chile [5] using the GHG Protocol standard found that 68% of total GHG emissions were from scope 3. Of these emission, student commuting and electricity consumption contributed 78%. It underscored the need for localized assessments and behavioral change

but does not propose any mitigating strategies. The study also notes a scarcity of research on GHG emissions from universities in developing countries, particularly in Latin America. The University of Wyoming's 2018 GHG inventory reported a 26.6% reduction in emissions compared to 2017, attributed to the adoption of greener fuel sources, decreased reliance on coal and natural gas, and a reduction in air travel [6]. Mendoza-Flores and Ortiz [7] found that commuting and electricity dominated emissions at UAM's Cuajimalpa campus, contributing 51% and 24% of total emissions, respectively. Var'on-Hoyos et al. [8] analyzed the Technological University of Pereira's 2017 carbon footprint, finding that scope 3 emissions, primarily from daily commuting and construction, accounted for 97% of 8,969 tons of CO_2 equivalent (tCO_2e). UTP had a lower carbon footprint per person compared to global universities due to not generating electricity or using heating, and the short commuting distances of its students and staff. Cano et al. [9] evaluated Universidad Nacional de Colombia's carbon footprint. They found scope 1, 2, and 3 emissions were 2.84% (418.63 tCO_2e), 14.03% (1,099.33 tCO_2e), and 83.13% (5,981.07 tCO_2e), respectively. The per capita emissions were 0.432 tCO_2e. Scope 3 emissions were highest due to extensive transportation. The studies did not explore strategies for reducing the significant emissions related to commuting, nor did they investigate the potential for using renewable energy sources. Clabeaux et al. [10] assessed Clemson University's GHG emissions, estimating 95,418 metric tons of CO_2. The study found that Scope 1 emissions made up 19%, Scope 2 emissions 40.57%, and scope 3 emissions 40.53%, with electricity consumption and transportation being major contributors. However, these studies did not identify opportunities for transitioning to cleaner energy sources or propose strategies to mitigate GHG emissions.

European universities have made notable progress in reducing GHG emissions, particularly through sustainable mobility initiatives and renewable energy adoption, though challenges remain in addressing material and waste-related emissions. Filimonau et al. [11] assessed Bournemouth University's carbon footprint during the COVID-19 lockdown, highlighting a 30% decrease from the previous year. Scope 3 emissions, mainly due to work from home, were 73%, while scope 1 and scope 2 were 6% and 21%, respectively. The per capita footprint was 0.16 tCO_2e. Study conducted by P'erez-Neira et al. [12] analyzed mobility at the University of León, Spain, using LCA, and highlighted the potential for significant GHG reductions through sustainable commuting, especially for distances within 0-6 km. The study suggested replacing cars with bicycles for trips under 4 km, which could reduce CO_2 emissions by 17% and increase cost savings by 14.8%. It also emphasized the role of universities in promoting sustainable mobility. Valls-Val and Bovea [13] found that the Universitat Jaume I's 2019 emissions were from scope 1 and scope 3, with no scope 2 emissions. Scope 3 emissions, from material use, waste, and transportation, were significant, but per capita emissions decreased to 0.3 tons per student, indicating progress in sustainability. However, the opportunity for more sustainable energy sources remains unknown. A 2019 study of the University of Oulu in Finland [14] used a hybrid methodology combining Environmentally Extended Input-Output Analysis (EEIOA) and Life Cycle Assessment (LCA), reporting a total footprint of 19,072 tCO_2e. The largest contributors were district heating (40%), procurement activities (22.16%), restaurant services, business travel, and commuting, reflecting the importance of local climate and operational factors in emission profiles. Complementing these findings, a recent study at the Technical University of Cluj-Napoca (TUCN) in Romania [15] assessed GHG emissions for 2022 and 2023. Using the GHG Protocol and One Click LCA software, the study found that stationary fuel use, primarily natural gas, remained the largest contributor to emissions, accounting for approximately 62.7% and 66.9% in 2022 and 2023, respectively. Scope 3 emissions declined from 28.3% in 2022 to 23.1% in 2023.

Universities in the Middle East face unique sustainability challenges, with high electricity consumption for cooling systems being a major driver of emissions, as seen in the case of the American University of Sharjah. A study conducted in Middle East by Samara et al. [16] found that for 2018-2019, the American University of Sharjah's carbon footprint, based on the GHG Protocol, had scope 2 emissions at 61.12% due to high electricity use. Scope 3 emissions were 38.51%, and scope 1 was 0.37%. The per capita emissions were 15.7 tCO_2e, lower than the UAE average of 20.8 tCO_2e. However, they are beginning to integrate carbon accounting practices, often aligning with global standards and customizing them for regional contexts. For example, Kamel et al. [17] evaluated the carbon footprint of the College of Maritime Transport and Technology in Alexandria, Egypt, for the 2022–2023 academic year. Using a tailored methodology based on the GHG Protocol Corporate Standards aligned with ISO 14064-1 and incorporating a LCA for selected indirect emissions, the total GHG emissions were determined to be 2,393 tCO_2e. Purchased electricity emerged as the largest source at 38.4%, followed by commuting at 25.6%.

In Africa, carbon assessments in HEIs are emerging, often constrained by data limitations but increasingly aligned with global standards like the GHG Protocol. Admas et al. [18] assessed the carbon footprint of Debre Markos University's main campus in Ethiopia using the GHG Protocol standard. In 2022, total emissions were 6,420 tCO_2e. Scope 1 sources, including vehicle fuel, diesel generators, land use, and firewood cooking, made up 64%. Scope 2 emissions from electricity and water use contributed 5%, while Scope 3 emissions from commuting, wastewater treatment, and paper use accounted for 31%. The per-student footprint was about 0.23 tCO_2e.

Universities across Asia exhibit diverse GHG emission patterns, with high electricity consumption from fossil-fuel-based grids being a dominant contributor, as seen in studies from Malaysia, Thailand, India, Indonesia, and Taiwan. In Malaysia, Yazdani et al. [19] assessed the carbon footprint at University Technology Malaysia (UTM) using the GHG Protocol model. The study found that purchased electricity, due to high energy demand and reliance on fossil fuels, accounted for 84% of total emissions. Solid waste disposal and transportation contributed 3.5% and 12.5%, respectively. In 2011, UTM emitted 57,576 $MtCO_2$, with each person's contribution being 2.1 $MtCO_2$. Kandananond [20] studied GHG emissions from only scope 1 and 2 GHG at Valaya Alongkorn Rajabhat University in Thailand. The Faculty of Industrial Technology and the Faculty of Agriculture Technology had the highest scope 2 emissions due to their significant electricity use in the laboratories, making them major sources of GHG emissions at the institution. In related effort, Thanatrakolsri and Sirithian [21] examined GHG emissions at Thammasat University's Lampang Campus in Thailand using the GHG Protocol Corporate Standard developed by WRI and WBCSD. The annual average emissions from 2019 to 2022 were 1051.70, 778.28, 558.64, and 1034.53 tCO_2e, respectively. Under normal operating conditions, electricity procurement emerged as the largest emission source, contributing about 78.55 percent of total emissions. Other indirect emissions made up 12.92 percent, while direct emissions contributed 8.53 percent. In the baseline year, the estimated average emissions were 0.43 metric tons per person, 0.50 per student, and 3.74 per employee. Sangwan et al. [22] assessed GHG emissions at the Pilani campus of BITS, finding scope 1, 2, and 3 emissions at 1.0%, 50.1%, and 48.9%, respectively, with per capita emissions of 4.65 tCO_2e. The high scope 2 emissions were mainly due to extensive air-conditioning. However, the study was not comprehensive, as it did not rely entirely on a single approach but rather combined elements of both the GHG Protocol and LCA methodology. Extending this line of research, Prasad and Deswal [23] assessed the carbon footprint of the National Institute of Technology Kurukshetra using an integrated approach that combined the GHG Protocol and the LCA method. For the 2022–23 fiscal year, the institute's total carbon footprint was 11,254.08 tCO_2e. Indirect emissions from electricity use accounted for 87 percent of the total, while other indirect emissions, mainly from commuting, contributed 12.17 percent. Direct emissions represented just 0.83 percent. The per student carbon footprint was 2.24 tCO_2e. Based on the GHG Protocol, Ridhosari and Rahman [24] assessed Universitas Pertamina's carbon emissions. They found that electricity use was the main source, accounting for 92.3% of total emissions, followed by transportation (6.66%) and trash generation (1.04%). The annual per capita emissions were estimated at 0.52 $MtCO_2$. Mustafa et al. [25] reported that NED University of Engineering & Technology's emissions were 7.4% from scope 1, 7% from Scope 2, and 85.6% from scope 3, mainly due to commuting and waste management, with per capita emissions at 1.79 tCO_2e. Wang [26] analyzed energy use at Taiwanese universities, noting that electricity accounted for 92% of total energy consumption and that public universities emitted more GHGs than private ones due to stricter energy controls in profit-driven private institutions. An empirical study at Zhejiang Ocean University in China applied the emission factor methodology combined with survey-based data from students and staff, estimating total emissions at 33,900 tCO_2e in 2021. Scope 3 emissions, primarily from personal energy use, water, and paper consumption, contributed the most at 54.68%, highlighting the significance of indirect activities in HEI emission profiles in East Asia [27]. However, these studies do not address or propose any mitigation strategies to reduce CO_2 emissions.

From these studies, the importance of localized assessments and strategies for reducing GHG emissions within HEIs is clearly evident. Building upon these insights, our work aims to fill the critical gap regarding GHG emissions in a Bangladeshi university context. This study not only evaluates the emissions profile of a HEI in a resource-constrained developing country — characterized by limited financial resources, minimal access to advanced monitoring infrastructure, constrained institutional capacity, and budget for sustainability — but also details practical interventions to reduce emissions. To do this, this study presents a scenario based mitigation framework with three tiers: Conventional, Incremental, and Comprehensive. Each scenario outlines progressively ambitious interventions across Scopes 1, 2, and 3. While the case study is based on a single institution, the proposed mitigation strategies are designed to be broadly applicable to universities across Bangladesh. In particular, this paper adds a unique perspective by (1) evaluating a resource-constrained environment, (2) identifying major emissions contributors, (3) proposing cleaner energy pathways, and (4) developing targeted mitigation strategies. The objectives of this study are as follows:

1) To assess the GHG emissions of a Bangladeshi university using the GHG Protocol Corporate Standard, and categorizing emissions into scopes 1, 2 and 3.
2) To identify and analyze the major emission sources within each scope and quantify their contributions to the university's total GHG emissions.
3) To calculate the institution's per capita carbon footprint and benchmark its emissions profile against those of other HEIs globally.
4) To recommend practical mitigation strategies tailored to the institutional context that effectively reduce GHG emissions.

By considering the unique challenges and opportunities in Bangladesh, this study aims to contribute to reducing the university's carbon footprint and act as a catalyst for sustainable practices within universities in Bangladesh. This aligns with the SDGs by promoting a sense of

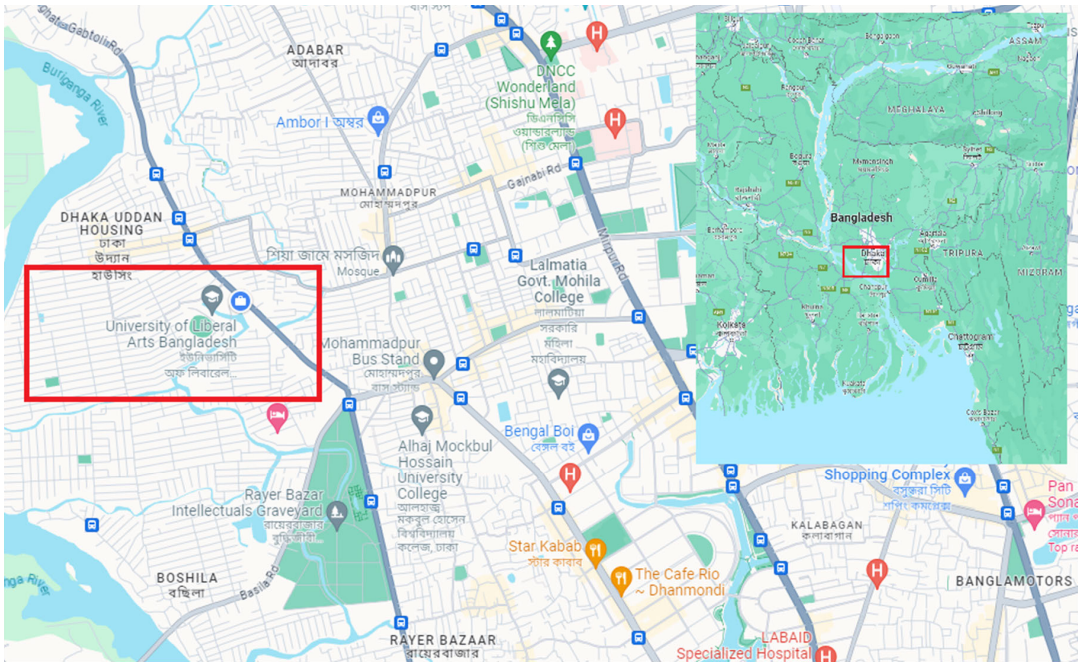


FIGURE 1. Location of the campus of the University of Liberal Arts Bangladesh (Dhaka, Bangladesh).

responsibility toward a more sustainable and climate-resilient future.

The remainder of this paper is organized as follows: Section II presents a description of the university of this case study and detailed explanation of the methodology. In Section III, the results of the emissions assessment are presented by scope, source of emission, type of GHG emission, and carbon footprint per capita. The results are then discussed and compared with GHG emissions from various universities worldwide, followed by proposed mitigation strategies and recommendations for reducing GHG emissions. Finally, Section IV concludes the paper by summarizing the key findings and suggesting directions for future research.

II. MATERIALS AND METHODS

This section outlines the context of the case-study university in Dhaka, the rationale for using the GHG Protocol Corporate Standard, and the step-by-step process for measuring and reducing emissions. We also highlight assumptions and excluded sources to ensure transparency.

A. STUDY AREA

This study assesses GHG emissions and proposes mitigation strategies for University of Liberal Arts Bangladesh (ULAB) in Dhaka, Bangladesh. The campus is located just 3.5 km from the city center as shown in Fig. 1.

ULAB, a private university in Dhaka, serves as a representative case study for urban HEIs in developing countries. Established in 2004, ULAB has since become a prominent HEI in the country. Located in a warm temperate climate with average daily temperatures ranging from 24 °C to 36 °C, the campus requires extensive use of air conditioning, which significantly contributes to its emissions.

During the 2022-23 academic year, as reported by its administration, offered 6 undergraduate and 4 master's programs, with an enrollment of 5000 students, mostly at the undergraduate level, The university employed around 350 individuals, including teachers, administrative and support staff. In addition to teaching, the university focuses on research and development, and there are 12 research centers. The university campus has four buildings covering a total construction area of 11,210 square meters and a total of 57,000 square meters of surface area encompassing classrooms, laboratories, library, offices, green spaces, roads, parking, playground, and basketball court.

One of the guiding principles for this university is "*Sustainable Development: Employing sustainable practices and respect for human rights, in order to become responsible leaders of the future*". This principle influences the university's initiatives and practices, reinforcing its commitment to sustainability across all operations. Consequently, the university aims to not only excel academically but demonstrate a strong commitment to SDGs. This dedication to sustainability makes it an ideal choice for this research work to assess the GHG emission and mitigation strategies for a private university in Bangladesh.

B. GREENHOUSE GAS PROTOCOL

Globally recognized protocols for GHG emissions measurement and reporting include several well-established frameworks, each offering unique strengths. ISO 14064, developed

by the international organization for standardization (ISO), provides a detailed, auditable approach suitable for organizations requiring rigorous third-party verification [28]. The IPCC Guidelines, created by Intergovernmental Panel on Climate Change (IPCC), commonly used for national-level inventories and offer sector-specific methodologies for calculating emissions [29], [30]. The publicly available specification (PAS) 2050, published by British Standards Institutions, focuses on product-level carbon foot printing, providing valuable insights into lifecycle emissions [31], [32].

While these frameworks offer valuable approaches The GHG Protocol Corporate Standard, developed by WRI and WBCSD, stands out as the most suitable due to several factors. It is one of the most widely adopted frameworks for GHG accounting and management [2] and is globally recognized as the first comprehensive framework for evaluating carbon footprints at the organizational level [33]. It enables organizations to identify, measure, and report their GHG emissions. It supports three distinct approaches to calculating emissions [34]. The protocol is further strengthened by five guiding principles—relevance, completeness, consistency, transparency, and accuracy—which ensure that emissions data are comprehensive, precise, and aligned with organizational goals. By following these principles, organizations can conduct accurate, consistent, and transparent assessments of their carbon footprints [35].

The GHG Protocol Corporate Standard classifies emissions into three primary scopes, each offering distinct insights into an organization's carbon footprint. Scope 1 includes direct emissions originating from sources owned or controlled by the organization. Examples of Scope 1 emissions encompass fuel combustion in generators, heating systems, and emissions from university-owned vehicles. These emissions are integral to the organization's operational activities and are considered direct emissions. Within Scope 2 category, indirect emissions result from the generation of purchased electricity. It reflects the organization's impact on GHG emissions through its energy consumption. By assessing Scope 2 emissions, organizations gain a comprehensive view of their electricity-related carbon footprint. Scope 3 emissions encompass other indirect emissions resulting from organizational activities, but from sources not owned or controlled by the organization. Examples of Scope 3 emissions include those stemming from road transportation, supply chains, and commuting by employees and students. These emissions often extend beyond an organization's immediate control but provide valuable insights into broader sustainability efforts.

The GHG Protocol is particularly well-suited for universities due to their multifaceted operations, involving research, teaching, and a diverse range of facilities. This framework allows for the inclusion of all relevant emissions sources. By adopting the GHG Protocol, universities can not only accurately measure and manage their carbon footprint but also engage diverse stakeholders, including students, faculty, and the surrounding community, in climate action initiatives. This transparent approach supports the achievement of campus sustainability goals, such as carbon neutrality, while providing a valuable learning opportunity for the entire university community. Therefore, this research adopts the GHG Protocol Corporate Standard for its comprehensive nature, flexibility, and broad applicability as well as its reliability and widespread acceptance in the global community. Its structured approach enables precise measurement, reporting, and mitigation of emissions data at ULAB, ensuring alignment with international best practices. Furthermore, its compatibility with frameworks like ISO 14064 enhances the credibility and depth of this study, facilitating detailed reporting and third-party verification. The decision to use the GHG Protocol underscores a commitment to standardized, actionable methodologies in addressing climate change and advancing sustainability efforts.

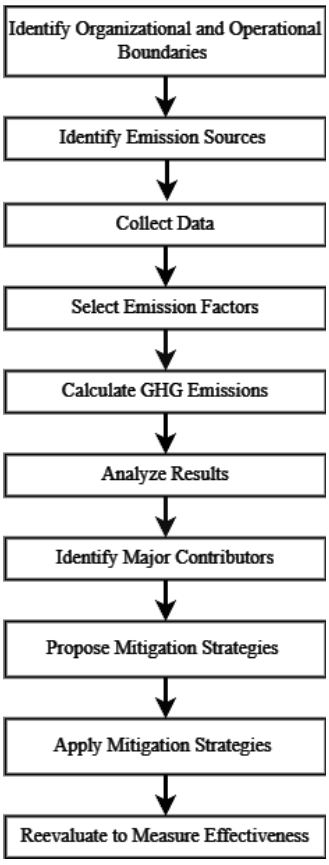


FIGURE 2. Overview of the proposed methodology.

C. METHODOLOGY

This study employs a systematic approach to comprehensively assess GHG emissions at a university in Bangladesh, as depicted in Fig. 2. The methodology also includes steps to identify major contributors and consequently recommend potential mitigation strategies.

The process begins by defining the organizational and operational boundaries, which is crucial for outlining the scope of the assessment. Once the boundaries are defined, the next step is to identify all relevant emission sources within the university according to the GHG Protocol Corporate Standard. This involves collecting quantitative data on all relevant emission sources from university records, such as bills, logs, and reports, as well as through surveys. Ensuring the accuracy and reliability of the collected data is important.

Once the data is collected, appropriate emission factors are selected to convert the data into GHG emissions. These emission factors are sourced from reputable databases, such as those provided by the IPCC, local environmental agencies and existing literature. The emission factors are applied to the collected data to calculate the total GHG emissions, expressed in terms of carbon dioxide equivalent (CO_2e).

The calculated emissions are then analyzed to determine major contributors to the university's overall GHG footprint. Major contributors to GHG emissions are identified to focus mitigation efforts on the most vital areas. Based on the findings, the study proposes specific mitigation strategies designed to the university's unique context and needs.

The final step in this methodology involves applying the proposed mitigation strategies and conducting a re-evaluation to measure their effectiveness. This step, however, will be part of a future study, as this work is the first such assessment for this university. Analyzing these results over the year would allow us to identify trends and patterns. By demonstrating the effectiveness of this structured methodology, the study aims to provide a robust framework for assessing GHG emissions and proposing mitigation strategies to strength the sustainability practices of any university.

1) ORGANIZATIONAL AND OPERATIONAL BOUNDARIES

It is important to recognize the boundaries and limitations when calculating the GHG emissions and carbon footprint. The organizational boundaries for emission calculation were the university main campus; emissions from a few research centers outside the main campus were not considered. The operational boundaries were also defined. GHG emissions from air or land travel by faculty members attending conferences and workshops, student field trips, transport of various supplies, stationary, recycling and waste management, and infrastructure and construction activities were not included in the calculation.

D. EMISSION SOURCES AND DATA COLLECTION

To properly assess the GHG emissions at the university, it was necessary to identify the sources of these emissions and categorize them. This comprehensive data collection approach, summarized in Fig. 3, ensured an accurate assessment of GHG emissions at the university, enabling us to quantify its carbon footprint accurately. These insights help to adopt informed and effective mitigation strategies in line with the GHG Protocol Corporate Standard. It is important to note that air conditioners contribute to multiple scopes under the GHG Protocol. Specifically, the electricity consumed by air conditioning systems is categorized under Scope 2 as an indirect emission from purchased electricity, whereas refrigerant leakage associated with their operation constitutes a direct fugitive emission under Scope 1. This dual contribution has been appropriately reflected in our classification to ensure accuracy and transparency in the emissions assessment.

TABLE 1. Fuel consumption of generators.

Month	Diesel (Liter)
January	0.00
February	800.00
March	800.00
April	0.00
May	800.00
June	0.00
July	366.97
August	1045.87
September	2900.00
October	0.00
November	1600.00
December	800.00
Total	9112.84

TABLE 2. Fuel consumption in cafeteria.

Cylinder Size (kg)	Monthly Quantity	Gas Used (kg/year)
35	20	8400
12	30	4320
Total		12720

TABLE 3. Fuel consumption of institutional vehicles.

Month	Octane (Liter)	Diesel (Liter)
January	738.00	220.92
February	718.77	309.36
March	668.85	1717.89
April	736.12	2115.23
May	752.77	1950.92
June	66.08	1700.18
July	500.04	1510.55
August	848.29	1868.50
September	871.08	1848.87
October	979.15	1854.36
November	964.46	1988.90
December	749.62	2043.10
Total	8593.22	19128.78

1) SCOPE 1

Under this category, the direct emissions from sources owned or controlled by the university are considered. It includes emissions from fuel consumption in generators to produce electricity, fuel usage in the cafeteria for cooking, and emissions resulting from university-owned vehicles. Fugitive emissions, the unintentional and undesirable emission, leakage, or discharge of CFC (chlorofluorocarbons) and HFC (hydrofluorocarbons) gases from air conditioners on campus, also contribute to scope 1 emissions These emissions are integral to the direct operational activities of the university and are categorized as scope 1 emissions.

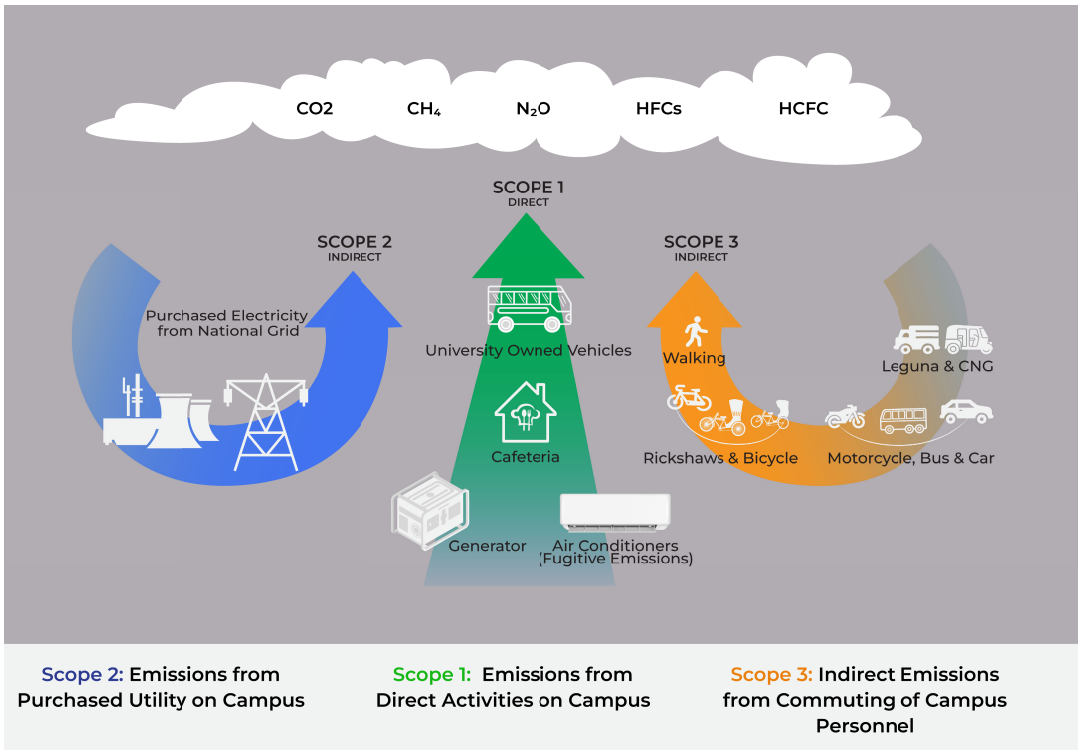


FIGURE 3. Sources of GHG emission from scopes 1, 2 and 3 at ULAB.

TABLE 4. On campus electricity consumption.

Month	Off Peak Hour (kWh)	Peak Hour (kWh)
January	12936.00	3828.00
February	17196.00	4080.00
March	67716.00	9528.00
April	54372.00	5076.00
May	60816.00	7536.00
June	61200.00	9660.00
July	61200.00	3600.00
August	53124.00	7392.00
September	50856.00	6528.00
October	48708.00	7260.00
November	47256.00	6132.00
December	30036.00	5376.00
Total electricity consumed at ULAB (kWh)		641412.00
Electricity generated (kWh) for ULAB considering 11% loss in Bangladesh		711967.32

a: FUEL CONSUMPTION

Information regarding fuel consumption in generators and cafeteria operations was obtained directly from the university administration. The university uses diesel in its generator. Table 1 shows that 9112.84 liters of diesel were consumed in the year 2022 by the generators.

The university cafeteria uses liquefied petroleum gas (LPG) for cooking. Table 2 shows that 12720 kilograms of LPG were consumed in the year 2022 for cooking in the cafeteria.

b: STUDENTS, TEACHERS AND STAFF COMMUTING BY INSTITUTIONAL VEHICLES

The university provides transportation by bus and minibus for the students, teachers, and staff to commute to and from the campus. The emissions from these vehicles are included in scope 1 because the fuel is purchased and used by the university. The total distance traveled by these vehicles was not available. However, we had access to the total amount of fuel purchased in a month, as outlined in Table 3, and used this information to calculate the GHG emissions.

Table 3 shows that 8593.22 liters of octane and 19128.78 liters of diesel were consumed in the year 2022 by the institutional bus and minibus.

c: FUGITIVE EMISSIONS FROM AIR CONDITIONERS (AC)

The air conditioners (AC) on the campus use R410A (50% HFC-32, 50% HFC-125) and R-22 (HCFC-22) gases as refrigerant. According to the data provided by the university administration, a total of 136 kg of R410A and 163 kg of R-22 refrigerant gases were refilled in the year 2022.

2) SCOPE 2

This category encompasses indirect emissions associated with the university's energy consumption, specifically from purchased electricity. This includes electricity used for lighting, air conditioners, computers, laboratory equipment, lifts, and other campus operations. It should be noted that the operational status (i.e., on/off) of individual equipment was not monitored. Scope 2 emissions were estimated based on the total electricity consumption recorded by campus utility meters over the reporting period.

TABLE 5. Types of transport used and average distance travelled daily by university personnel.

Transport	Teachers	Staff	Students	All	Avg. Distance (km/day)
Walking	0	8	33	41	2.51
Bicycle	2	2	10	14	5.54
Rickshaw	9	23	166	198	6.08
Battery Auto-rickshaw	7	6	46	59	6.51
Leguna	0	2	12	14	12.15
Motorcycle	8	7	48	63	16.72
Public Bus	1	2	114	117	29.02
CNG Auto-rickshaw	1	4	37	42	15.72
Car	14	7	46	67	19.86
ULAB Transport	4	6	35	45	–
Multiple Transport	7	16	120	143	18.09
Total	53	83	667	803	–

a: ELECTRICITY CONSUMPTION

The electricity consumption data on the campus were obtained from the bills given to the university by the electric power supply company. Table 4 shows that the off-peak hours, peak hours, and total electricity consumption were 565416 kWh, 75996 kWh and 641412 kWh respectively. Considering the total loss of 11% in the power sector of Bangladesh, as mentioned in the 2021-2022 annual report by Bangladesh Power Development Board (BPDB) [36], the energy generated for ULAB was 711967.32 kWh.

3) SCOPE 3

This category includes indirect emissions from teachers, staff and students commuting to the campus.

a: STUDENTS, TEACHERS AND STAFF COMMUTING BY PERSONAL OR RENTED VEHICLES

To evaluate scope 3 emissions from personal or rented vehicles used by teachers, staff, and students, we used online surveys using Google Forms and face-to-face interviews. The survey achieved 803 responses, which is about 23% of the university commuters, and provided valuable data on distance travelled and types of vehicles used to commute to and from campus. Table 5 shows the commuting behavior of the university community.

To calculate the GHG emission for scope 3, the commuters using ULAB transport have been omitted since the GHG emission related to this has already been included in scope 1. While motorcycles and CNG autorickshaws use gasoline and compressed natural gas (CNG) respectively, cars and, in some cases, buses use different types of fuel. Most cars run on gasoline, CNG, or a combination of both. Given these diverse fuel types used by vehicles, the survey included questions about fuel types to calculate GHG emissions accurately.

TABLE 6. Emission factors for various scope 1 sources.

Emission Source	Fuel	CO_2	CH_4	N_2O	Unit
Institutional Vehicle	Gasoline	2.272	0.00032782	0.00001967	kg/l
	Diesel (transport)	2.676	0.0003612	0.00002167	kg/l
Cooking in Cafeteria	LPG	1.612	0.00037976	0.00000255	kg/l
Electricity (Generator)	Diesel (stationary)	2.676	0.0003612	0.00002167	kg/l
Air conditioner	R410A	2255.5	–	–	$kgCO_2e/kg$
	R-22	1960	–	–	$kgCO_2e/kg$

E. EMISSION FACTORS AND EQUATIONS

One of the most crucial steps in calculating the GHG emission is to determine the associated emission factor (EF) for each source.

1) SCOPE 1

Since the EFs for diesel used in generator to produce electricity and institutional buses for transportation, for gasoline used in institutional car and minibus, LPG used for cooking, particularly for Bangladesh could not be found in literature. The EFs used in this section have been obtained from "Emission Factors from Cross-Sector Tools" by the GHG Protocol [37]. The EFs for R410A (50% HFC-32, 50% HFC-125) and R-22 (HCFC-22) refrigerant gases used in air-conditioners are obtained from the 7SM Earth's Energy Budget, Climate Feedbacks and Climate Sensitivity Supplementary Material [38] of the Sixth Assessment Report (AR6) of [39]. The information is summarized in Table 6.

The data on total distance traveled by institutional vehicles was unavailable. Therefore, an alternative approach to estimate GHG emissions, depending on total fuel consumption, was utilized in this work. Under this methodology, the total fuel purchased for consumption in a year, as shown in Table 3, is multiplied by the appropriate EF presented in Table 6 as shown in the following equation (1):

$$\mathrm{GHG_{veh}} = \mathrm{EF_{gas}} \cdot \mathrm{Gas_{purch}} + \mathrm{EF_{diesel}} \cdot \mathrm{Diesel_{purch}} \quad (1)$$

For the generators, the annual GHG emissions were calculated by multiplying the EF for diesel with the quantity of diesel purchased in a year, presented in Table 1, as shown in the following equation (2):

$$\mathrm{GHG_{generators}} = \mathrm{EF_{diesel}} \cdot \mathrm{Diesel_{purch}} \quad (2)$$

Similarly, the GHG emissions from LPG used in cafeteria was calculated using the following equation (3):

$$\mathrm{GHG_{cafeteria}} = \mathrm{EF_{LPG}} \cdot \mathrm{LPG_{purch}} \quad (3)$$

The EFs for R410A (50% HFC-32, 50% HFC-125) and R-22 (HCFC-22) refrigerant gases used in air-conditioners (AC) are given in CO_2e, the GHG emissions were calculated based

TABLE 7. Electricity generation in Bangladesh and weighted average EF of GHG in scope 2.

Source	Percentage of power generated from each source $P_{gen\,(\%)}$	Average EF of each source (gCO_2e/kWh)	EF of each source based on $P_{gen\,(\%)}$ (CO_2e/kWh)
Gas	55.06%	533.17	293.56
Coal	6.24%	942.33	58.80
Oil	26.71%	773.80	206.68
Solar PV	0.38%	65.05	0.25
Hydro	0.87%	8.22	0.07
Diesel	1.73%	266.70	4.61
Power Import	9.01%	820.00	73.88
$EF_{weighted\ average}$			637.86

on sales-based approach of GHG protocol using the following equation (4):

$$GHG_{AC} = EF_{HFC/HCFC} \cdot (HFC/HCFC)_{purch} \quad (4)$$

2) SCOPE 2

To determine the EF associated with the electricity consumption, it is important to first understand the energy sources of Bangladesh's power grids. Fossil fuels (natural gas, oil, coal and diesel) are used to produce 89.74% of the electricity in Bangladesh according to the 2021-22 Annual Report by BPDB [36].

Firstly, similar to Akib et al. [40] and Rasheduzzaman et al. [41], the EF for GHG in CO_2e — comprising CO_2, CH_4, and N_2O — is determined based on the percentage of power produced ($EF_{gen.,source}\%$) from each source.This is calculated by multiplying the EF of a source (EF_{source}) with the percentage of total electricity it produced ($P_{gen.,source}\%$), using the data from the 2021-2022 annual report by BPDB [36], as shown in the following equation (5):

$$EF_{gen,\,source\%} = EF_{source} \cdot P_{gen,\,source\%} \quad (5)$$

Then, the weighted average EF ($EF_{wt.\,avg.}$) from these various sources on the basis of percentage of power generated is calculated utilizing the following equation (6).

$$EF_{wt\,avg} = \sum_{i=1}^{m} \left(EF_{gen,\,source\%}\right)^{(i)} \quad (6)$$

Here, m is the number of sources. As shown in Table 7, the weighted average EF for GHG ($EF_{wt.\,avg.}$) was calculated to be 637.86 gCO_2e/kWh, based on the current mix of electricity generation sources in Bangladesh. These sources include natural gas, oil, coal, solar, hydro, diecel and imported power from India. The EFs of diesel and imported power were obtained from [42] and [43] respectively.

Finally, the total GHG emission in CO_2e is determined by multiplying the weighted average EF with the total electricity purchased by the university (Table 4) as shown in the following equation (7).

$$GHG\,(CO_2e) = EF_{wt\,avg} \cdot Electricity_{purch} \quad (7)$$

3) SCOPE 3

Under Bangladesh Air Pollution Studies (BAPS) project, the Department of Environment (DoE) in Bangladesh, as per the report by Randall et al. [44], prepared the emission inventory for Dhaka and Chittagong investigating various sectors such as industry, road traffic, non-road traffic, agriculture, urban, and fossil fuel used in energy and gas processing for emissions of PM_{10}, $PM_{2.5}$, NO_x, SO_x, and CO. However, they calculated GHG emissions from road traffic by utilizing EFs derived in the context of India, as outlined by Baidya and Borken-Kleefeld [45] and The Automotive Research Association of India [46]. In contrast, Wadud [47] relied on literature focused on South Asian countries and made necessary modifications for Dhaka city to determine the EFs from various types of vehicles and fuels. The EFs for various vehicles and fuel types are primarily taken from Wadud [47]. The EFs of N_2O for various vehicles and fuel types were obtained from the EFs for GHG inventories by United States Environmental Protection Agency [48].

Since on average there are 52 passengers on a public bus in Dhaka city, the GHG emissions per passenger were calculated using equation (8):

$$GHG_{per\ passenger} = \frac{GHG_{per\ bus}}{N_{passengers}} \quad (8)$$

The GHG emissions per passenger for leguna, one of the unique vehicles in Dhaka, uses diesel as fuel and typically carries 12 passengers, was also calculated similarly.

If vehicles use more than one type of fuel, then we need to know the percentage of vehicles exclusively using only one type of fuel and the percentage of vehicles using multiple types of fuel. In addition, among the vehicles using multiple types of fuel, it is also essential to know the percentage of usage of each fuel. In this work, the EFs for mixed fuels are calculated considering all the factors. First of all, for the cars which consume both gasoline and CNG ($EF_{gas.\&CNG}$), the EF of CO_2, assuming that 20% gasoline and 80% CNG, was calculated to be 241.2 gCO_2/km using the following equation (9):

$$EF_{gas\&CNG} = 20\% \times EF_{gas} + 80\% \times EF_{CNG} \quad (9)$$

According to the survey, of all the cars commuting to the university, 32 cars (66.67%) use only gasoline, 6 cars (12.50%) use only CNG, 2 cars (4.17%) use only diesel, and 8 cars (16.67%) use both gasoline and CNG. The weighted average EF of CO_2 for a sedan car ($EF_{sedan,wt.\,avg}$) was calculated to be 255.68 gCO_2/km using the following equation (10):

$$EF_{sedan,\,wt\,avg} = \sum_{i=1}^{n} (EF_{fuel} \cdot \%\text{fuel use})^{(i)} \quad (10)$$

Here, n is the number of fuel types.

The weighted average EF of a SUV, ridesharing and microbus are also calculated in a similar way. Finally, the weighted average EF of a car was calculated to be

TABLE 8. Emission factors for various types of cars in scope 3.

Vehicle Type	Fuel Type	No of vehicles	Emission Factors (g/km)		
			CO_2	CH_4	N_2O
Sedan	Gasoline	32	258	0.14	0.00087
	CNG	6	237	2.53	0.00250
	Diesel	2	332.5	0.14	0.01190
	Gasoline-CNG (this work)	8	241.2	2.05	0.00217
	$E.F._{sedan,weighted\ average}$		255.68	0.76	0.00175
SUV	Gasoline	5	331	0.14	0.00075
	CNG	1	332.5	0.14	0.00310
	Gasoline-CNG	2	304	2.53	0.00193
	$E.F._{SUV,weighted\ average}$		324.44	0.74	0.00134
Ride-sharing	CNG	8	237	2.53	0.00250
Microbus	Gasoline	2	665	0.06	0.00075
	CNG	1	736	8.46	0.00310
	$E.F._{microbus,weighted\ average}$		685.33	2.86	0.00153
$E.F._{car,weighted\ average}$		67	280.90	1.06	0.00178

280.90 gCO_2/km based on the percentage of each type of vehicles as shown in the following equation (11):

$$EF_{car,\ wt\ avg} = \sum_{i=1}^{p} \left(EF_{veh,\ wt\ avg} \cdot \%veh\right)^{(i)} \tag{11}$$

Here, p is the number of vehicle types. The results were summarised in Table 8.

One of the unique vehicles in Dhaka city is the battery-operated auto-rickshaw. These vehicles use electricity to charge the batteries. Rahman et al. [49] determined that battery-operated autorickshaws consume 0.088 kWh per kilometer, while Awal et al. [50] calculated the value to be 0.0714 kWh per kilometer. Using the average value of these two studies, the electricity consumption in this work was determined to be 0.0797 kWh per kilometer. Considering the EF per kilowatt-hour associated with electricity generation in the Bangladesh power grid, the EF per kilometer for battery-operated autorickshaws was calculated using the following equation (12):

$$EF_{bat\text{-}op\ auto} = Electricity_{consumed} \cdot EF_{electricity} \tag{12}$$

According to the survey, 18.87% of members of the university community use multiple modes of transport for commuting to and from the university. However the survey provided partial information regarding all the modes of transportation for individuals, and the distance travelled by each transportation. Since the average daily distance travelled using multiple modes of transport was 18.09 km, it was assumed that a person would travel around 6 km using a rickshaw or battery-operated auto-rickshaw and 12.09 km using bus or leguna. We also assumed that 20% of people took rickshaw and 80% people took battery-powered auto-rickshaw to travel the 6 km distance; and 20% people took bus and 80% people took leguna to travel the 12.09 km distance. The weighted average EFs per kilometer were then calculated

TABLE 9. Emission factors for various types of transports in scope 3.

Vehicle Type	Fuel Type	No of vehicles	Emission Factors (g/km)		
			CO_2	CH_4	N_2O
Walking	N/A	41	0	0	0
Bicycle	N/A	14	0	0	0
Rickshaw	N/A	198	0	0	0
Car	Mixed	67	280.90	1.06	0.00178
Bus	Diesel	117	17.06	0.0012	0.00052
Leguna	Diesel	14	27.71	0.0117	0.00223
Battery-operated auto-rickshaw	Electricity	59	50.84		
CNG auto-rickshaw	CNG	42	75	1.41	0.00250
Motorcycle	Gasoline	63	40	0.08	0.00516
Multiple transport	Mixed	173	30.58	0.01	0.00126

using the following equation (13):

$$EF_{mult\ tr} = \frac{\sum_{i=1}^{q} (\%people \cdot EF_{tr} \cdot D_{tr})^{(i)}}{Total\ Dist_{mul\ tr}} \tag{13}$$

The EFs of all these various vehicles have been summarized in Table 9. It can be noted that the emissions related to ULAB transport are excluded here since those are already accounted in Scope 1.

In order to calculate the annual emissions of a GHG for all the vehicles in a category, we should first determine the annual GHG emissions by one vehicle in that category. First, the average distance travelled annually (Avg. $DTA_{one\ veh.}$) by one vehicle of a specific category commuting to the university was calculated by multiplying the average distance travelled daily by one vehicle (Avg. $DTD_{one\ veh.}$) with the number of the academic calendar days (ACD), which was 294 days in the year 2022, as shown in the following equation (14):

$$Avg\ DTA_{one\ veh} = Avg\ DTD_{one\ veh} \cdot ACD \tag{14}$$

Then the annual GHG emissions by one vehicle in that category was calculated by multiplying the average annual distance travelled by one vehicle in that category by the EF of the vehicle as shown in the following equation (15).

$$GHG_{one\ veh} = Avg\ DTA_{one\ veh} \cdot EF_{GHG} \tag{15}$$

Finally, the annual emissions of a GHG for all the vehicles in that category was calculated by multiplying the annual GHG emissions by one vehicle in that category by the total number of vehicles in that category as shown in the following equation (16):

$$GHG_{all\ veh\ in\ a\ cat} = GHG_{one\ veh} \cdot N_{veh} \tag{16}$$

For example, according to the survey (Table 5), the average distance travelled daily by a person to commute to and from the university using a motorcycle (mc) was 16.72 km/day, which results in average distance travelled annualy to be 4915.68 km/year. Since motorcycles in Bangladesh only use gasoline, which has an EF of 40 gCO_2/km, the annual

TABLE 10. Global warming potential (GWP_{100}) of various GHG.

GHG	GWP-100
CO_2	1
CH_4	27.9
N_2O	273
HFCs	2255.5
HCFC	1960

CO_2 emissions by a motorcycle was calculated to be 0.1966272 tCO_2/year using the following equation (17):

$$CO_{2,\text{one mc}} = EF_{CO_2\text{-gas}} \cdot \text{Avg DTA}_{\text{one mc}} \quad (17)$$

From the survey (Table 5), it has been observed that a total number university personnel who use motorcycles for commuting is 63. The total CO_2 emissions by all the motorcycles in a year were calculated to be 12.39 tCO_2/year using the following equation (18):

$$CO_{2,\text{all mcs}} = CO_{2,\text{one mc}} \cdot N_{\text{mcs}} \quad (18)$$

In a similar way, the annual GHG emissions from all types of vehicles used for commuting to and from ULAB by the university community personnel are calculated and presented in the next section. The CH_4 and N_2O emissions are also calculated.

The total GHG in this category was calculated by adding the GHG emission from all the individuals using multiple modes of transportation. Since the calculation is based on the number of people participated in the survey; the GHG emission per person was first calculated and then multiplied by the total number of personnel at the university to determine the total scope 3 emissions.

F. GLOBAL WARMING POTENTIAL AND EQUIVALENT CO_2 EMISSIONS

The global warming potential (GWP) is a metric used to compare the global warming impacts of different gases. It quantifies how much energy the emissions of 1 ton of a gas will absorb over a specified period, relative to 1 ton of carbon dioxide (CO_2). A higher GWP indicates a greater warming effect compared to CO_2 over that time frame, which is typically 100 years. The global warming potential of various GHG over 100 years (GWP-100), according to the Sixth Assessment Report (AR6) of IPCC [39], are presented in Table 10.

Once the emissions of each GHG, i.e., CO_2, CH_4, N_2O, are calculated for all three scopes, the GHG emissions are converted to CO_2e using the following equation (19).

$$CO_2e = GHG_{\text{emission}} \cdot GWP_{100} \quad (19)$$

G. PER CAPITA CARBON FOOTPRINT

The per capita carbon footprint was calculated in tCO_2e by dividing the total GHG emissions from all scopes (GHG_{total}) by the total campus population ($\text{Population}_{\text{campus}}$), which includes students, faculty and staff present during the reporting period, as shown in the following equation (20):

$$\text{Per Capita Carbon Footprint} = \frac{GHG_{\text{total}}}{\text{Population}_{\text{campus}}} \quad (20)$$

This metric enables comparison across institutions of varying sizes and helps contextualize total emissions on a per-person basis.

TABLE 11. Annual GHG and CO_2e emissions in scope 1 at ULAB in the year 2022-2023.

Source	CO_2 (t/year)	CH_4 (t/year)	N_2O (t/year)	HFC (tCO_2e)	HCFC (tCO_2e)	CO_2e per source (t/year)
Uni. Veh	70.71	0.00973	0.00058	-	-	71.14
Generators	24.39	0.00329	0.00020	-	-	24.53
Cafeteria	20.50	0.00483	0.0000325	-	-	20.65
AC	-	-	-	306.75	319.48	626.23
CO_2e/GHG (t/year)	115.60	0.4980	0.2221	306.75	319.48	742.55 (total)
% of total CO_2e/GHG	15.57	0.07	0.03	41.31	43.02	

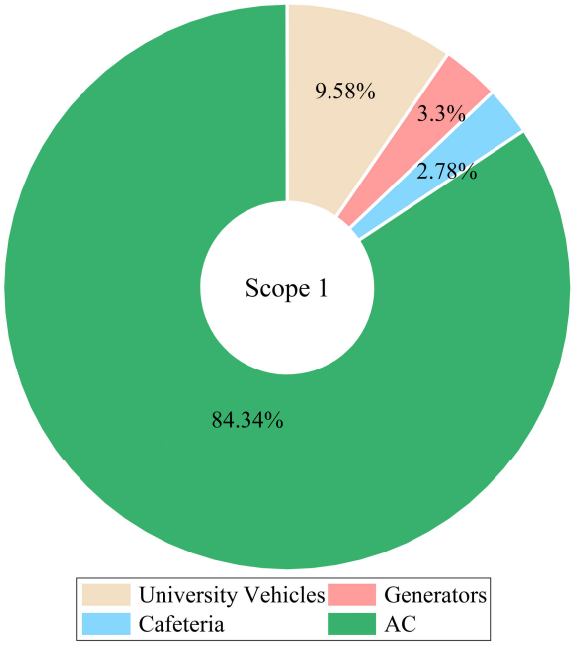


FIGURE 4. Percentage of GHG emissions from various sources in scope 1 for the year 2022-2023 at ULAB.

III. RESULTS AND DISCUSSION

A. GREENHOUSE GAS EMISSIONS AT ULAB

1) SCOPE 1

Table 11 and Fig. 4 show the GHG, i.e., CO_2, CH_4, N_2O, and CO_2e, emissions and percentage of emissions from various sources in scope 1 at ULAB.

Scope 1 emissions at ULAB primarily originate from university vehicles, generators, cooking in the cafeteria, and fugitive emissions from air conditioning (AC) units. The total annual GHG emissions from these sources amount to 742.55 tCO_2e. University vehicles contribute 71.14 tCO_2e,

TABLE 12. Total annual carbon dioxide equivalent (tCO_2e) from scope 2 (2022-2023) at ULAB.

Energy Consumption at ULAB (kWh)	Considering 11% loss, Energy Generated for ULAB (kWh)	GHG Emission Factor (gCO_2e/kWh)	GHG Emission (tCO_2e/year)
641412.00	711967.32	637.86	454.14

which accounts for 9.58% of scope 1 emissions. Generators contribute 24.53 tCO_2e, representing 3.30%, while the cafeteria adds 20.65 tCO_2e, equating to 2.78%. The largest share of scope 1 emissions comes from AC units, contributing a substantial 626.23 tCO_2e annually, which represents 84.33% of scope 1 emissions as shown in Fig. 4. The dominance of AC units in scope 1 emissions likely stems from the continuous cooling requirements in Dhaka's warm climateand suggests a potential area for improvement.

2) SCOPE 2

Table 12 shows the tCO_2e in scope 2 at ULAB. Scope 2 emissions, resulting from the consumption of purchased electricity, were calculated to be 454.14 tCO_2e per year. This represents 18.18% of the total emissions, which is a notable amount of emissions. This data suggests that reduction efforts would be beneficial and might impact the overall carbon footprint.

3) SCOPE 3

The scope 3 emissions originate from indirect emissions from teachers, staff and students commuting to the campus. The distribution of transport modes and average distance travelled daily among all the university personnel at ULAB who participated in the survey, which is around 14% of the university community, has been presented in Table 5.

As shown in Fig. 5, rickshaws were the most commonly used mode of transport among all ULAB personnel (26%), followed by multiple modes of transport (19%) and public buses (15%). On the other hand, bicycle and leguna were the least used transports. The data showed that the primary choice of transport of students and staff is rickshaw (26% and 30%, respectively), whereas cars are the most used transport (29%) by the faculty members. Those living near the university typically prefer rickshaws, while those residing farther away rely on cars, buses, or other public transportation options. This trend is significant, as cars, with their higher EFs per kilometer, contribute substantially to Scope 3 emissions.

Table 13 shows the annual GHG, i.e., CO_2, CH_4, and N_2O and CO_2e per year from various transport modes in scope 3 at ULAB.

Scope 3 emissions are responsible for the largest portion of ULAB's total GHG emissions, totaling 1302.14 tCO_2e annually. These emissions primarily arise from the commuting of university personnel. Cars are the primary source of these emissions, contributing 110.05 tCO_2e per year, which accounts for 59.62% of transport-related scope 3 emissions

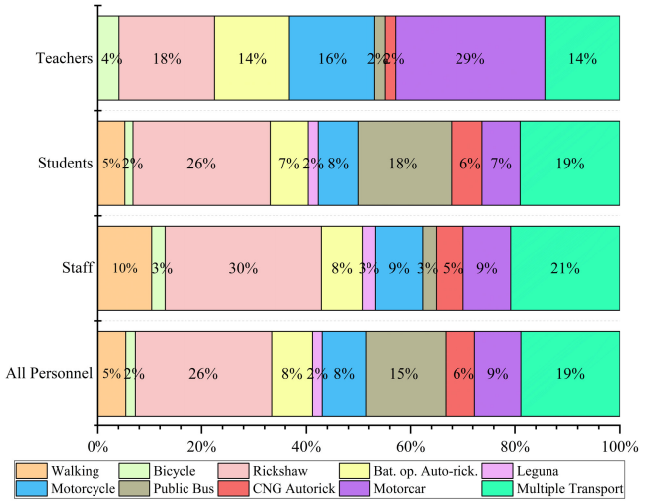


FIGURE 5. Percentage of transportation modes for the year 2022-2023 at ULAB.

TABLE 13. Annual GHG and CO_2e emissions in scope 3 at ULAB in the year 2022-2023.

Transport Mode	CO_2 (t/year)	CH_4 (t/year)	N_2O (t/year)	CO_2e / Transport Mode (t/year)
Walking	0	0.00000000	0.00E+00	0
Bicycle	0	0.00000000	0.00E+00	0
Rickshaw	0	0.00000000	0.00E+00	0
Bat.-op. auto-rickshaw	5.75	0.00000000	0.00E+00	5.75
Leguna	1.39	0.00000229	5.77E-10	1.39
Motorcycle	12.39	0.00031752	2.70E-08	12.40
Public Bus	17.03	0.00001580	9.31E-09	17.03
CNG auto-rickshaw	14.56	0.00248724	5.82E-09	14.63
Car	109.91	0.00476209	1.05E-08	110.05
Multiple Transport	23.27	0.00013071	3.41E-08	23.27
CO_2e/GHG (t/year)	184.27	0.21526642	2.38E-05	184.49 (total)
% of total tCO_2e/GHG	99.88%	0.12%	0.01%	

as shown in Fig. 6. Public buses contribute 17.03 tCO_2e (9.23%), motorcycles add 12.40 tCO_2e (6.72%), battery-operated auto-rickshaws contribute 5.75 tCO_2e (3.11%), and CNG auto-rickshaws contribute 14.63 tCO_2e (7.93%), while multiple transport modes add 23.27 tCO_2e (12.61%). The high usage of cars highlights the need for ULAB to promote more sustainable transportation options, such as additional transportation services for employees.

It must be noted that although 803 people participated in the survey, scope 3 calculations were performed for 758 people, excluding those who used ULAB transport, as their emissions are already accounted for in the scope 1 calculation. Therefore, to determine the total scope 3 emissions at ULAB, the emissions per person in scope 3 were calculated first and then multiplied by 5350, the total number

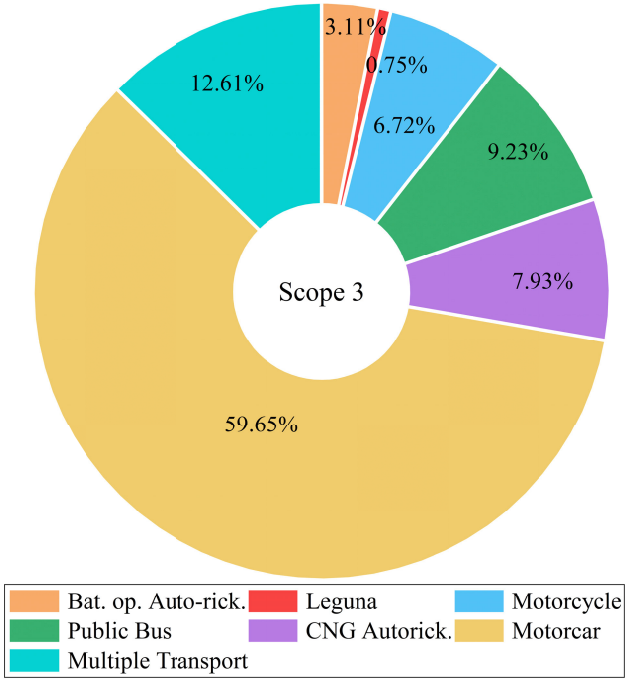


FIGURE 6. Percentage of GHG emissions from various sources in scope 3 for the year 2022-2023 at ULAB.

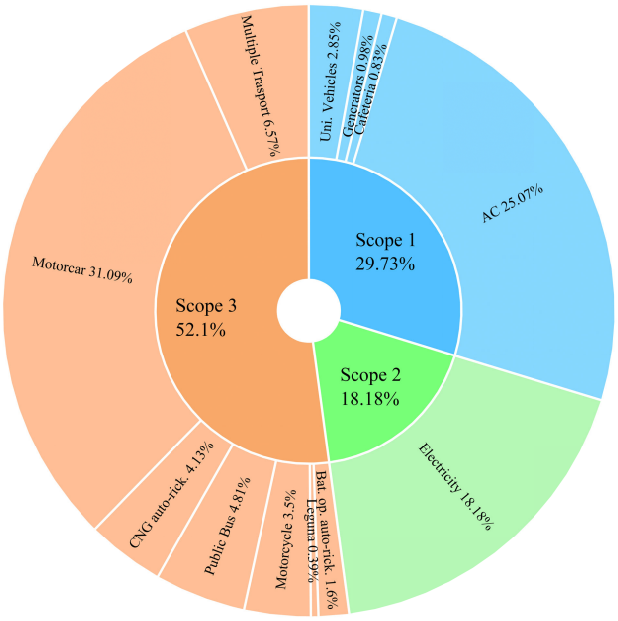


FIGURE 7. Percentage of GHG emissions from various sources and scopes for the year 2022-2023 at ULAB.

TABLE 14. Total annual emissions from scopes 1, 2, and 3, and carbon footprint per capita for the year 2022-2023 at ULAB.

Source	CO_2e (t/year)
Scope 1	742.55
Scope 2	454.14
Scope 3	1302.14
Total	2498.80
Total Personnel at ULAB	5350
Carbon footprint per capita (tCO_2e/year)	0.4671

of ULAB personnel as shown in equation (21):

$$\text{Scope } 3_{\text{total}} = \left(\frac{184.49}{758}\right) \cdot 5350 = 1302.14\,\text{tCO}_2\text{e} \quad (21)$$

4) TOTAL GHG EMISSIONS

Table 14 shows the total tCO_2e from scopes 1, 2, and 3, their contribution in percentage, and the carbon footprint per capita at ULAB.

The total annual GHG emissions for ULAB are calculated to be 2498.80 tCO_2e and the carbon footprint per capita is 0.4671 tCO_2e. Scope 1 emissions contribute 742.55 tCO_2e, accounting for 29.73% of the total as shown in Fig. 7. Scope 2 emissions are the smallest fraction, at 454.14 tCO_2e, making up 18.18% of total emissions. Scope 3 emissions are the largest contributor, with 1302.14 tCO_2e, representing 52.10% of the total emissions. It can be clearly seen from Fig. 7 that emissions from AC, electricity and cars contribute significantly, 25.07%, 18.18% and 31.09% respectively, to the total GHG emissions at ULAB.

B. COMPARISON OF GHG EMISSIONS WITH VARIOUS UNIVERSITIES

Table 15 presents an overview of carbon emissions across various universities, highlighting significant differences in the contributions of scopes 1, 2, and 3, as well as the per capita carbon footprint.

The findings reported in these studies showed notable differences in scope 1 emissions among universities. The National Institute of Technology (NIT) Kurukshetra in India [23] reported scope 1 emission of only 0.83% of its total carbon footprint for the fiscal year 2022–23. Var'on-Hoyos et al. [8] reported an emission of 0.9% at the University of Pereira in Colombia. Kandananond [20] reported a slightly higher scope 1 emission of 1.1% at Valaya Alongkorn Rajabhat University in Thailand. The University of Oulu in Finland [14] had Scope 1 emissions at only 1.68% of its total in 2019. Zhejiang Ocean University in China [27] reported an even smaller proportion, with Scope 1 emissions making up 0.99% of its total in 2021, based on data gathered through site visits to their logistics department. The College of Maritime Transport and Technology (CMTT) in Egypt [17] also had a low Scope 1 contribution, at 1.6% for the 2022-2023 academic year. On the other hand, Thammasat University's Lampang Campus in Thailand [21] had a moderate scope 1 emissions contributing 8.53% of its total emissions. The studies conducted by Clabeaux et al. [10] at Clemson University in the United States and Valls-Val and Bovea [13] at the Universitat Jaume I in Spain showed a considerably higher scope 1 emissions, 18.9% and 26.76% of the total emissions profile respectively. In this work, ULAB also showed a notably high Scope 1 emission of

TABLE 15. Comparison of GHG emissions and carbon footprint per capita at various universities.

University	Year of Study	Pub. Year	Population	Method Used	Contribution of each scope (%)			Per capita carbon footprint (tCO_2e)
					1	2	3	
The American University of Sharjah, UAE [16]	2018-2019	2022	5122	GHG Protocol	0.37	61.12	38.51	15.7
NED University, Pakistan [25]	2017	2022	12000	GHG Protocol	7.4	7	85.6	1.79
Universitat Jaume I, Spain [13]	2019	2022	15884	GHG Protocol	26.76	0.00	73.23	0.3
Bournemouth University, England [11]	2020	2021	17893	GHG Protocol	6	21	73	0.16
University of Pereira, Colombia [8]	2017	2021	20067	GHG Protocol	0.9	2.5	96.6	0.4
Universidad Nacional de Colombia, Colombia [9]	2019	2023	16783	UNE-ISO & GHG Protocol	2.84	14.03	83.13	0.432
Universitas Pertamina,Indonesia [24]	2017-2018	2023	2621	WARM	6.66	92.3	1.04	0.52
Clemson University, USA [10]	2014	2020	21857	HLCA	18.9	40.57	40.53	4.4
Birla Institute of Technology and Science Pilani, India [22]	2014-2015	2018	5699	GHG Protocol	1.1	50	48.9	4.65
Valaya Alongkorn Rajabhat University, Thailand [20]	2017	2017	10365	–	1.87	98.12	–	0.064
University Technology Malaysia, Malaysia [19]	2011	2013	7284	WARM	12.5	84	3.5	2.1
Technical University of Cluj-Napoca, Romania [15]	2022 & 2023	2025	18000	GHG Protocol	63 & 63	8.7 & 13.9	28.3 & 23.1	0.37 & 0.362
University of Oulu, Finland [14]	2019	2023	16900	EEIOA & LCA	1.68	40.82	57.50	1.129
Zhejiang Ocean University , China [27]	2021	2024	17522	LCA	0.99	44.33	54.68	1.027
College of Maritime Transport and Technology, Egypt [17]	2022-2023	2024	2693	GHG Protocol	1.6	38.4	60	2.67
Debre Markos University, Ethiopia [18]	2022-2023	2025	30449	GHG Protocol	64	5	31	0.2108
National Institute of Technology, India [23]	2022-2023	2024	5027	GHG Protocol & LCA	0.83	87	12.17	2.24
Thammasat University, Thailand [21]	2019-2022	2024	2402 (2019), 389 (2020), 386 (2021), 2333 (2022)	GHG Protocol	8.53	78.55	12.92	0.43
University of Liberal Arts Bangladesh [this work]	2022-2023	–	5350	GHG Protocol	29.73	18.18	52.10	0.467

29.73%. In contrast, some universities reported extremely high Scope 1 contributions. For the Technical University of Cluj-Napoca (TUCN) in Romania [15], scope 1 emissions were a significant contributor to their total carbon footprint, accounting for 63% in both 2022 and 2023. This high percentage was primarily due to stationary fuel use, with natural gas being the largest contributor, making up approximately 62.7% of total emissions in 2022 and increasing to 66.9% in 2023. Debre Markos University (DMU) in Ethiopia [18] reported a substantial 64% of its total carbon footprint from Scope 1 emissions in 2022. These direct sources included fuel consumption by university vehicles, backup diesel generators, land use, and cooking with firewood. In contrast, several other universities showed much lower contributions from Scope 1. The highest scope 1 contributions were observed at Debre Markos University (DMU) in Ethiopia (64%), and the American University of Sharah in UAE [16] reported the lowest scope 1 contribution of only 0.37%. ULAB's significantly higher contribution of 29.73% suggests a high dependance on direct emissions, particularly from the fugitive emission from the air-conditioners on campus.

Scope 2 emissions also showed significant variations, primarily highlighting the various levels of dependencies on purchased electricity on campus. Valaya Alongkorn Rajabhat University in Thailand reported highest percentage of scope 2 emissions at 98.12% [20], followed by Universitas Pertamina in Indonesia at 92.3% [24]. Similarly, a large contribution of 84% scope 2 emissions was reported at the University of Technology Malaysia [19]. For the National Institute of Technology (NIT) Kurukshetra in India [23], indirect emissions from electricity use were also the most significant contributor, accounting for 87% of its total emissions for the fiscal year 2022–23. Likewise, at Thammasat University's Lampang Campus in Thailand [21], electricity procurement was identified as the largest emission source under normal operating conditions, contributing about 78.55% of total emissions. Samara et al. [16] reported 61.12% emission in scope 2 in American University of Sharjah (AUS), and Clabeaux et al. [10] at Clemson University in the United States reported 40.57% scope 1 emission. These emissions were primarily produced by the heavy electricity usage in their classrooms and laboratories on their campus. Several other universities also showed moderate Scope 2 contributions. The University of Oulu in Finland [14] reported Scope 2 emissions constituting 40.82% of its total carbon footprint in 2019, with approximately 40% coming specifically from district heating. Zhejiang Ocean University in China [27] had Scope 2 emissions accounting for 44.33% of its total in 2021. The College of Maritime Transport and Technology (CMTT) in Egypt [17] also found purchased electricity to be its primary contributor, making up 38.4% of its total GHG emissions for 2022–2023. In contrast to these high and moderate scope 2 emitters, several institutions reported relatively lower Scope 2 contributions. The Technical University of Cluj-Napoca (TUCN) in Romania [15] had Scope 2 emissions of 8.7% in 2022, which increased slightly to 13.9% in 2023. Debre Markos University (DMU) in Ethiopia [18] also had a small Scope 2 share of just 5% of its total emissions in 2022, which included indirect emissions from both purchased electricity

and water consumption. In this study, ULAB in Bangladesh reported a moderate scope 2 emission of 18.18%. In can be noted that, according to [13], the Universitat Jaume I in Spain reported no scope 2 emissions, a remarkable reduction from previous years where emissions were significantly higher, 41.27% and 54.18% reported in 2016 and 2017 respectively. This reduction is attributed to the efficient implementation of action plans aimed at curbing emissions within the institution's operational framework.

Scope 3 emissions, including all other indirect emissions, primarily from commuting and waste management, were the most significant contributors for many universities, and also showed significant variations across universities. The University of Pereira has the highest contribution from scope 3 emissions at 96.6% [8] and NED University in Pakistan followed closely with 85.6% [25]. Bournemouth University in England [11] reported 73% scope 3 emissions, which similar to the 73.23% emission reported at the Universitat Jaume I in Spain [13]. The College of Maritime Transport and Technology (CMTT) in Egypt [17] also found Scope 3 to be the largest emissions category, comprising 60% of its total for the 2022–2023 academic year, with commuting alone accounting for 25.6%. Similarly, Zhejiang Ocean University in China [27] reported Scope 3 emissions at 54.68% of its total in 2021, which included personal energy consumption, water, and paper usage. The University of Oulu in Finland [14] also reported Scope 3 as the largest component of its carbon footprint in 2019, contributing 57.50% of the total GHG emissions. Key contributors in this category included procurements (22.16%), restaurant services (12.87%), business travel (8.58%), and commuting (7.56%). Moderate Scope 3 emissions were observed at several universities. In this work, scope 3 emissions of 52.10% were reported at ULAB, similar to 48.9% scope 3 emissions at the Birla Institute of Technology and Science Pilani in India [22]. Debre Markos University (DMU) in Ethiopia [18] also reported a notable Scope 3 share of 31% in 2022, arising from activities such as wastewater treatment, staff and student commutes, and paper consumption. The Technical University of Cluj-Napoca (TUCN) in Romania [15] reported Scope 3 emissions of 28.3% in 2022, which decreased to 23.1% in 2023. The National Institute of Technology (NIT) Kurukshetra in India [23] had Scope 3 emissions of 12.17%, primarily from commuting during the 2022–2023 fiscal year. Thammasat University's Lampang Campus in Thailand [21] reported Scope 3 emissions of 12.92%. On the lower end, the University Technology Malaysia [19] and Universitas Pertamina in Indonesia [24] reported a negligible scope 3 emission of only 3.5% and 1.04% respectively, generated from waste management.

The population size at different universities did not show any significant impact on the per capita carbon footprint. Clemson University in the USA [10], with a population of 21,857, reported a per capita carbon footprint of 4.4 tCO_2e, whereas the University of Pereira in Colombia [8], with a similar population of 20,06, reported a per capita carbon footprint of 0.4 tCO_2e. Institutions with moderate populations also showed wide variation. The American University of Sharjah (AUS) in the UAE [16], with a population of 5,122, had the highest reported per capita footprint of 15.7 tCO_2e. In contrast, the National Institute of Technology (NIT) Kurukshetra in India [23], with a comparable population of 5,027, reported a much lower footprint of 2.24 tCO_2e. University Technology Malaysia in Malaysia [19], with a population of 7,284, reported a similar per capita carbon footprint of 2.1 tCO_2e. Strong contrasts have been also observed among institutions with very small populations. Universitas Pertamina in Indonesia [24], with a population of 2,621 and per capita carbon footprint of 0.52 tCO_2e, while the College of Maritime Transport and Technology (CMTT) in Egypt [17], with similar population of 2,693 people, reported a relatively high 2.67 tCO_2e.

The per capita carbon footprint varied widely among universities, influenced by local climate, energy consumption, commuting behaviors, and sustainability practices. The American University of Sharjah [16] had the highest per capita emission at 15.7 tCO_2e, largely due to heavy electricity use for lighting, heating, and cooling, though this is below the UAE national average of 20.8 tCO_2e. Birla Institute of Technology and Science Pilani in India [22] reported 4.65 tCO_2e per person, mainly due to extensive air conditioner use, while Clemson University in the USA [10] had a similar footprint of 4.4 tCO_2e from power and transportation. Similarly, the College of Maritime Transport and Technology in Egypt reported a substantial 2.67 tCO_2e per capita for 2022–2023 [17], and the National Institute of Technology Kurukshetra in India calculated 2.24 tCO_2e per student during the same fiscal year [23]. Moderate per capita emissions were observed at NED University in Pakistan [25] and University Technology in Malaysia [19], respectively 1.79 tCO_2e and 2.1 tCO_2e, driven by energy use and commuting. In Asia and Europe, Zhejiang Ocean University in China reported 1.027 tCO_2e per person [27], and the University of Oulu in Finland reported 1.129 tCO_2e per person [14], both reflecting institutional and regional energy practices. Low carbon footprints of 0.432 tCO_2e and 0.52 tCO_2e were reported by Universidad Nacional de Colombia [9] and Universitas Pertamina [24] respectively, due to renewable energy use and smaller operational sizes. Universities of Pereira in Colombia [8] and Jaume I in Spain [13] achieved reduced emissions through sustainability initiatives like photovoltaic (PV) electricity and upgraded systems. Valaya Alongkorn Rajabhat University [20] had the lowest per capita footprint at 0.064 tCO_2e, though scope 3 emissions were excluded. Bournemouth University's low footprint of 0.16 tCO_2e reflected the COVID-19 lockdown impact rather than actual policies [11]. Other universities also reported notably low per capita carbon footprints. Debre Markos University in Ethiopia recorded approximately 0.2108 tCO_2e, while the Technical University of Cluj-Napoca in Romania reported 0.378 tCO_2e in 2022, which declined

slightly to 0.362 tCO_2e in 2023 [15], [18]. Thammasat University's Lampang Campus in Thailand estimated an average of 0.43 tCO_2e per person in its 2019 baseline year [21]. This study calculated modest emissions of 0.4671 tCO_2e per capita for ULAB.

C. MITIGATION STRATEGIES AND RECOMMENDATIONS

This section presents a structured set of mitigation strategies and institutional recommendations aimed at reducing GHG emissions in Bangladeshi universities, including a scenario-based analysis applied to ULAB.

1) MITIGATION STRATEGIES FOR UNIVERSITIES

Tertiary education institutions worldwide are increasingly embedding sustainability practices at the heart of their organizational ethos. As highlighted in existing literature, the three primary components that shape the strategies for fostering a sustainable campus include knowledge dissemination, robust governance frameworks, and effective environmental management of operations [51]. Within this context, a set of mitigation strategies is proposed for universities in Bangladesh, with the principal objective of reducing institutional GHG emissions while nurturing a culture of sustainability.

a: ENERGY EFFICIENCY AND ADOPTION OF RENEWABLE ENERGY

Energy efficiency is a cornerstone of GHG mitigation. Universities can significantly reduce their energy consumption and GHG emissions by implementing policies and actions focused on improving energy efficiency, adopting renewable energy sources, promoting green purchasing, and encouraging sustainable landscaping [52]. Routine energy audits are essential to identify inefficiencies in university buildings and infrastructure. Energy-saving strategies such as installing energy-efficient heating, ventilation, and air conditioning (HVAC) systems, switching to light-emitting diode (LED) lighting, and using smart meters for monitoring and controlling energy use are highly effective [53].

Universities in Bangladesh can capitalize on the abundant sunshine by installing solar panels on campus rooftops. A study conducted at Ahsanullah University of Science and Technology (AUST) [54] demonstrated the technical and economic feasibility of integrating renewable energy sources into a university's energy infrastructure. The research focused on the design and analysis of a 256 kWp grid-connected PV system, intended to supply approximately 21% of the institution's total electrical load. The system adheres to established engineering and safety standards for PV adoption and incorporates a net energy metering framework, enabling surplus electricity to be fed back into the grid for credit. The analysis estimated a levelized cost of energy (LCOE) of BDT 2.39, indicating substantial cost-effectiveness. Moreover, the proposed system is projected to reduce CO_2 emissions by approximately 4,875 tons over its operational lifespan, underscoring the environmental benefits of renewable integration in HEIs. Establishing research facilities focused on renewable energy technologies and encouraging academics and students to work on innovative projects using renewable energies, specially solar and wind power, can further reduce dependency on fossil fuels [55].

b: ECO-FRIENDLY MOBILITY OPTIONS

Promoting eco-friendly mobility options can significantly reduce GHG emissions. Universities should establish secure and accessible bicycle routes and provide incentives and parking spaces for those opting for non-motorized transportation [56]. Collaborating with public transportation authorities to enhance access for university personnel, along with instituting carpooling programs and shuttle services, can reduce individual car trips and alleviate traffic congestion.

At the policy level, introducing public buses on university roads can provide reliable, large-capacity transport alternatives, thereby reducing the single use of motorcycles and private cars. In addition, universities can implement digital ride-matching platforms to support carpooling initiatives for students and staff traveling from similar locations. These institutional interventions not only reduce Scope 3 emissions but also promote a culture of shared, sustainable mobility across the campus community.

A technical study conducted at the University of Dhaka [57] proposed an innovative model for integrating solar energy with electric mobility to facilitate student transportation. The concept involves establishing a 9.9 kW PV system to power a dedicated charging station for battery-electric buses operating within the university's vicinity. While electric buses are not yet widely available in Bangladesh, the study explores the feasibility of locally assembling or importing a 14-seat electric bus to serve as a campus shuttle. The proposed vehicle configuration includes a battery capacity of 8.64 kWh, an energy consumption rate of 0.086 kWh/km, a driving range of 100 kilometers, and a top speed of 40 km/h. Although the initial investment cost is relatively high, the model presents a long-term opportunity to reduce emissions and promote sustainable transportation in academic institutions.

The bicycle is a forgotten and neglected mode of transport, even though it offers numerous individual, social, and environmental benefits over motorized alternatives. Several studies have investigated factors influencing the adoption of bicycles for commuting. For instance, Alfatha et al. [70] explored students' willingness to cycle in a university setting, utilizing a modified theory of planned behavior and identifying attitudes, subjective norms, perceived behavioral control, perceived barriers, and habit as key influences. Likewise, Schönbach et al. [59] outlined a systematic review protocol for school-based interventions to promote active school transportation by bicycle among children and adolescents, focusing on intervention strategies and effects. More recently, Márquez et al. [60] examined motivations for

workers to commute by bicycle in Bogota, integrating the theory of planned behavior and social identity into a Hybrid Modal Shift model to understand the impact of both hard and soft measures.

c: ECO-FRIENDLY CONSTRUCTION AND CAMPUS PLANNING

Adopting green building guidelines for new constructions and major campus upgrades is crucial. New buildings should be designed to meet user needs while incorporating elements such as green roofs, rainwater harvesting systems, sustainable materials, and energy-efficient designs. Given the significant GHG emissions from air conditioning, building designs should prioritize natural ventilation to reduce reliance on cooling systems. Creating green spaces by planting trees, establishing gardens, and developing green belts enhances biodiversity and provides natural cooling and recreational areas for the university community.

Bangladeshi universities hold substantial potential to reduce GHG emissions by adopting low-carbon construction materials and sustainable architectural practices. Fired clay bricks and ordinary Portland cement are carbon- and energy-intensive and contribute highly to the built environment's emissions. Through a transition to alternatives such as compressed stabilized earth blocks (CSEBs) and fly ash bricks, any building can be more CO_2 emission free. One study found that CSEBs can potentially reduce the carbon emission during the manufacturing to almost 178kg CO_2/tonnes [61]. Furthermore, the integration of passive design strategies—including natural ventilation, daylighting, reflective roofing, and rainwater harvesting—can significantly lower operational energy consumption. These practices not only align with Bangladesh's national climate action plans but also comply with the Green Building Guideline issued by the Sustainable and Renewable Energy Development Authority [62], offering a comprehensive framework for the development of environmentally responsible campuses.

d: SHIFT TO SUSTAINABLE BEHAVIOR

Enhancing the awareness of the people within HEIs about energy efficiency, decarbonization, and the transition toward sustainable practices is crucial for promoting meaningful behavioral change and supporting institutional sustainability goals. Recent research emphasizes the importance of understanding the behaviors and perceptions of campus communities. For example, Nguyen et al. [63] conducted a human-centered study exploring students' opinions and preferences regarding a plastic-free university campus, using student feedback to prioritize actionable strategies. Similarly, another study [64] surveyed lecturers, staff, and students to assess their views on energy efficiency measures, highlighting the need to raise awareness in three key areas: recognizing electricity resource depletion, promoting resource sharing to reduce consumption, and understanding the importance of preparing for future energy shortages. These findings demonstrate that targeted awareness-raising activities can significantly improve occupant knowledge and encourage active participation in creating a greener, decarbonized campus environment. To build on these insights, it is recommended that HEI administrations systematically conduct annual surveys among students, faculty, and staff to assess behavioral perceptions related to energy use and sustainability. Based on survey results, institutions should organize targeted seminars, workshops, and incentive programs designed to address identified gaps and motivate positive behavioral change.

To strengthen long-term impact, HEIs should integrate sustainability into the curriculum to ensure that students from diverse academic backgrounds are aware of its importance. Specialized education in green technology, sustainable development, and environmental science should address both technical aspects as well as social norms and behaviors [65]. Additionally, training programs for faculty members, staff, and students can enhance awareness and equip them with the skills necessary to implement sustainable practices effectively. Funding and incentivizing research on regional and global sustainability issues, and fostering partner ships with businesses, governments, and NGOs can translate research into practical solutions. Leadership programs should be developed to equip students with the knowledge and skills needed to champion sustainability. Supporting student-led projects and initiatives can enhance sustainability efforts both on campus and in the larger community.

2) RECOMMENDATIONS FOR ULAB

Building on this foundation, a scenario-based mitigation framework was developed to address the specific institutional context of ULAB, while remaining broadly applicable to other universities in Bangladesh. The ULAB case study illustrates how broad mitigation strategies can be effectively operationalized through context-specific scenario modeling at the institutional level.

The proposed framework consists of three implementation scenarios, structured in increasing levels of ambition and institutional commitment: the Conventional Scenario, the Incremental Scenario, and the Comprehensive Scenario.

- The Conventional Scenario serves as the business-as-usual baseline, assuming no adoption of mitigation measures. Under this model, university operations continue unchanged, with no major shifts in technology, behavior, or infrastructure—resulting in the continuation of current GHG emission levels.
- The Incremental Scenario introduces partial implementation of feasible strategies, such as limited adoption of renewable energy, gradual phasing in of low-emission transport options, and selective behavior change interventions. This scenario reflects a balanced approach, where environmental responsibility is pursued within the constraints of operational and financial feasibility.

TABLE 16. Scenario-Based GHG Mitigation Strategies for ULAB.

Mitigation Strategies	Conventional Scenario	Incremental Scenario	Comprehensive Scenario
Replacement of AC refrigerants	No replacement	50% replacement	100% replacement
Adoption of Renewable Energies	No integration	Installation of Rooftop Solar System (RSS)	Deployment of Floating Photovoltaic (FPV), combined with RSS
Transition from Fuel to Electric in University Vehicles	No transition	50% reduction in fossil fuel use with limited electric vehicle integration	100% transition of university vehicle fleet to electric vehicles
Additional Transport Services for Faculty/Staff	No additional services	Introduction of hybrid shuttles on select routes	Fully electric transport fleet

It is particularly relevant for institutions at an early stage of their sustainability transition.

- The Comprehensive Scenario represents full institutional commitment to sustainability, involving the widespread implementation of advanced mitigation technologies, complete replacement of high-emission systems, and systemic behavioral transformation. This scenario prioritizes maximum GHG reduction through coordinated action across energy, transportation, infrastructure, and campus culture, aligned with long-term climate goals.

Table 16 highlights which strategies are adopted in each scenario:

Although behavioral change is included in the recommend framework, its impact has not been quantitatively modeled in this study due to the absence of baseline behavioral data or formal assessments at ULAB.

a: REDUCING FUGITIVE EMISSIONS FROM AIRCONDITIONERS

At ULAB, scope 1 emissions, particularly from the fugitive emissions of the ACs, are notably high at 29.73%. This significant emission level results from the use of coolant gases R410A and R22 in the ACs. A viable strategy to mitigate these emissions is transitioning to R32, an HFC with lower global warming potential and improved energy efficiency [66].

The Incremental Scenario considers a 50% replacement of currently used refrigerant gases in air conditioners with R-32. The substitution of R-410A with R-32 results in a 35% emission reduction, while replacing R-22 with R-32 achieves a 32.72% reduction. Combined, this partial intervention leads to a 33.83% overall decrease in refrigerant-related fugitive emissions, significantly contributing to Scope 1 mitigation.

In contrast, the Comprehensive Scenario proposes a complete replacement (100%) of R-410A and R-22 gases with R-32. This full-scale replacement yields a 70% reduction when shifting from R-410A and 65.42% from R-22, leading to a total emission decrease of 67.67% in refrigerant use. This marks a major step toward achieving clean cooling and reducing Scope 1 emissions.

b: ADDRESSING TRANSPORTATION EMISSIONS

The second highest emission source identified is scope 3, specifically related to transportation of ULAB teachers, students, and staff. This issue can be mitigated through two primary strategies:

In the transportation sector, the Incremental Scenario envisions a 50% transition from fossil fuel-powered university vehicles to solar-charged electric vehicles (EVs). This shift is estimated to cut emissions from university fuel consumption by 37.62%, promoting cleaner mobility solutions. Under the Comprehensive Scenario, the university replaces 100% of its vehicles with EVs using solar-powered charging stations, producing a 47.47% reduction in GHG emissions from transportation. In both scenarios, the equivalent electricity required to replace gasoline and diesel consumption is considered to be 9.7 kWh and 10 kWh per liter, respectively [67], [68].

Additionally, instead of faculty relying on individual cars, a shared hybrid minibus system is proposed for 14 faculty members in Incremental Scenario, based on a travel pattern of 20 km per day using 80% CNG and 20% gasoline, achieving a 37% emission reduction for that group. In the Comprehensive Scenario, the faculty transportation model is upgraded from hybrid to shared electric minibuses. Even when charged using grid electricity from a coal-based mix, this intervention can reduce emissions by 59%, providing both ecological and logistical advantages over private vehicle use [69].

c: ADOPTING RENEWABLE ENERGY

The emissions from Scope 2 constitute 18.18% of ULAB's total emissions; therefore, the university can adopt renewable energy to reduce this emission and further increase its sustainability efforts.

To address Scope 2 emissions, the Incremental Scenario includes the installation of a 42 kWp rooftop solar system (RSS). Based on findings by Alfatha et al. [70], this system is expected to generate 66.225 MWh of electricity annually, accounting for 10.33% of ULAB's total electricity demand. This intervention alone could reduce Scope 2 emissions by 18.26%, showing both environmental and economic viability.

Expanding upon this, the Comprehensive Scenario integrates a hybrid solar energy system that combines the 42 kWp rooftop solar system with a 116.5 kWp floating photovoltaic (FPV) installation, as proposed by Jawad et al. [71]. The FPV system alone is capable of generating 169,000 kWh annually, and together, the combined solar systems would supply up to 36.67% of ULAB's electricity demand, resulting in a 39.58% reduction in Scope 2 emissions. This blended

approach enhances energy resilience while maximizing the use of renewable sources.

IV. CONCLUSION

This study provides a comprehensive assessment of GHG emissions at ULAB using the GHG Protocol Corporate Standard, revealing a total annual GHG emissions of 2498.80 tCO_2e. Scope 1 emissions - primarily from university vehicles, generators, cafeteria, and fugitive emissions from air conditioners - accounted for 29.73% (742.55 tCO_2e) of the total emissions. Scope 2 emissions, associated with purchased electricity, were the smallest contributor at 18.18% (445.14 tCO_2e). Scope 3 emissions, which are indirect emissions from commuting of university personnel, were the largest contributor at 52.10% (1,302.14 tCO_2e). The carbon footprint per capita is 0.4671 tCO_2e. The study identified that emissions from ACs, purchased electricity and cars used by the university personnel for commuting are major contributors to the university's carbon footprint.

To mitigate GHG emissions at ULAB, this study introduces a scenario based mitigation framework consisting of Conventional, Incremental, and Comprehensive Scenarios, each outlining progressively ambitious interventions across Scopes 1, 2, and 3. To address the high emissions from cooling systems, the study recommends replacing the current coolant gases R410A and R22 with more efficient and less harmful alternatives such as R32. Transportation emissions can be mitigated by expanding university transport services, and introducing hybrid or electric buses. Adopting renewable energy, like small-scale floating photovoltaic systems and rooftop solar installations, could further ULAB's sustainability efforts. This study also proposes several strategies for universities to reduce GHG emission. Implementing energy-saving strategies, such as energy-efficient HVAC systems, LED lighting, and smart meters, alongside installing solar panels, can significantly reduce energy consumption and GHG emissions. Promoting eco-friendly mobility options, like secure bicycle routes and enhanced public transportation access, along with eco-friendly construction practices, such as green building guidelines and natural ventilation designs, is crucial in reducing reliance on energy-intensive cooling systems. Integrating sustainability into the curriculum, supporting research, and student-led initiatives are essential to foster a sustainable campus culture.

Future research should focus on expanding the operational boundaries of the study and include GHG emissions from air and land travels, field trips, supply transports, waste management and infrastructure activities. Following up after implementing the recommendations and encouraging interdisciplinary research will be crucial for addressing the complex issue of GHG emissions. This approach can lead to innovative solutions and holistic strategies for sustainability, contributing to long-term environmental resilience in the face of climate change. Additionally, future studies should assess the broader carbon footprint of food consumption at ULAB, including emissions from food production, procurement, and waste. Conducting a life-cycle analysis of food procurement and waste management would provide a more comprehensive understanding of the university's sustainability impact and inform policies for reducing food-related carbon emissions. While behavioral change is included as a component in the mitigation framework, its impact could not be quantitatively assessed in this study due to the lack of baseline behavioral data at ULAB. Future efforts should focus on collecting such data through surveys or pilot programs to better understand and evaluate the role of community actions in emission reduction. Air quality, particularly in terms of exposure to pollutants like $PM_{2.5}$ and PM_{10}, is another important aspect of campus sustainability. While not addressed in the present study, which followed the GHG Protocol framework, this remains a valuable area for future research.

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